Like, Comment, Get Scammed: Characterizing Comment Scams on Media Platforms

NDSS 2024 RESEARCH PAPER ANALYSIS

Introduction

- Study by Li et al. (NDSS 2024)
- Longitudinal analysis of comment scams on YouTube
- First comprehensive research on comment scams
- My goal: Evaluating the solution and related research

Problem context

- YouTube: 400 hours of content uploaded daily, 4.5 million views/minute
- Spam comments have evolved into sophisticated scams
- Current YouTube filter is insufficient
- Scammers use advanced tactics to deceive users like comments or replies, enticing users to contact them through messages
- They also solicit a chance to win a gift or investment opportunities
- Example: “TextMe on WhatsApp (555)-5555” but with special characters
- Scammers apply multiple tactics to evade platform restrictions

Research Methodology

- 6-month monitoring infrastructure: Crawl module: hourly captures through YouTube API
- Storage module: custom database
- Analysis module: three specialized filters
- Monitored Channels: 20
- Cooking / Sports / Finance etc.
- Captured comments: Million
- Filtered scam comments: 2.34% of total comments

Research Infrastructure

- Builded a reliable infrastructure monitoring YouTube comments
  - Monitored past and new videos in popular YouTube channels
  - Periodically took snapshots of comment sections
- Designed heuristic filters to identify scam comments
- Used a three-part system designed to monitor YouTube comments:
  - Crawl Module: Hourly snapshots to capture comment dynamics
  - Storage Module: Database for comments and metadata
  - Analysis Module: Three specialized filters (text, image, time-based)

Detection Filters

- Text-based Filter: Identified visually similar symbols (VSS) to bypass detection
- Image-based Filter: Identified channel owner impersonation
- Time-based Filter: Detected scripted conversations (often within 15 seconds)
- Found 94.14% of scams with text filter, 13.63% with image filter, 5.4% with time filter

Comment Scam Features

- Graphical - Scammers apply similar profile images to impersonate channel owners
  - Difficult to distinguish in the view of inexperienced users
  - Perceptual hashing to compare with channel owners
- Temporal - Scammers fabricate story and use multiple accounts to post them at the same time
  - Multiple accounts used
  - Most difficult to detect

Key Findings

- 206,306 fraudulent comments identified (2.34% of total)
- ~10,000 unique accounts posting ~1,140 comments daily
- Coordinated campaigns using hundreds of accounts
- Only 31.42% of scammer accounts deactivated within 6 months

Key Findings - Scam Typology

- Investment scams: promises of crypto/trading profits
- Prize scams: users must pay "shipping fees"
- Financial impact: $1M-$2M stolen (confirmed through blockchain analysis)
- Not only finance channels are targeted (cooking, politics channels too)

Key Findings - Scammer Interraction

- Explored scammer tactics and payment channels
- Platforms used: WhatsApp and Telegram
- Pretended to be unaware users (prospective victims) who are inexperienced with cryptocurrency / trading
- Answered scammers' questions with polite/positive attitude
- Collected chat text records from scammers
  - Chat history - WhatsApp and Telegram
  - Websites
  - Payment channels

Key Findings - Scammer Tactics

- Channel targeting: Finance channels most targeted (72%), but cooking and politics also affected
- Impersonation: 13.63% used channel owner's profile image
- Conversation patterns: Scripted dialogues with multiple accounts
- Evasion techniques: Unicode character substitution, account rotation

Comparison with Prior Research

- More comprehensive than previous work on YouTube spam (Alberto et al., Oh et al.)
  - Alberto et al. (2015): YouTube spam filtering (focused on detection, not analysis)
  - Oh et al. (2021): ensemble machine learning models for spam
  - Bartoletti et al.: study of general crypto scams (not connected to YouTube)
  - Costin et al.: analysis of phone numbers in cybercrime
- Unique contributions:
  - Longitudinal monitoring of major channels
  - Multi-filter detection combining textual, visual, and temporal patterns
  - Direct interaction with scammers to understand tactics
  - Quantification of financial damages through blockchain analysis

Conclusion & Implications

- Comment scams are a significant threat
- Platforms require much more sophisticated detection strategies
- Importance of combining textual, visual, and temporal analysis
- Future directions: improving real-time detection and educating users

Thank you!

ADDAPTED FROM [NDSS-24] LIKE, COMMENT, GET SCAMMED

AND ALSO FROM MY LITTLE RESEARCH ON THIS PAPER